Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/23/2006
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Forced Flow Ignition and Flame Spread Test (FIST)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ferpello@me.berkeley.edu	**Fax:**	FY 510 642-6163
PI Organization Type:	UNIVERSITY	**Phone:**	510 642-6554
Organization Name:	University of California, Berkeley		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	6105A Etcheverry Hall		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
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	The objective of the program is to develop a new flammability apparatus and a bench-scale flammability test, FIST (Forced Ignition and Spread Test), that reproduces the environment that is expected in a manned space-based facility, i.e., microgravity and low velocity gas flows of variable oxygen concentration. The FIST would subsequently be used to obtain “flammability diagrams” of practical materials (plastics, composites, etc.) that are expected in these facilities. The research could eventually provide NASA with a test methodology to complement the existing protocol and provide a more comprehensive assessment of material flammability for space applications. Theoretical modeling, an extensive normal gravity data bank, and a few validation space experiments will support the testing methodology. This protocol could lead to better determination and ranking of the fire hazard characteristics of potential materials to be used in space-based environments, which are necessary criteria for an adequate fire protection design for space-based facilities. The FIST apparatus is similar to the ASTM E-1321 LIFT apparatus, except the fuel sample is mounted in small-scale wind tunnel so that a low-velocity oxidizer flow of specified oxygen concentration can be forced over the sample		

Task Description:	surface during the test. The oxidizer (air or a N2/O2 mixture of prescribed oxygen concentration) is forced through a rectangular duct at a known velocity. The duct is 56 cm in length, 10 cm high, and 12.7 cm wide. Its two side walls are clear Pyrex and its bottom surface is an insulating substrate (mica). A 30 mm by 30 mm test specimen is embedded flush in the substrate so that the boundary layer is not disrupted. Two radiant heaters are used to irradiate the sample with an approximately uniform heat flux of known intensity. Ignition is induced at fuel sample's trailing edge with an electrically heated Kanthal wire coil that acts as a pilot. Measured quantities for a particular material include the time to ignition as a function of ambient oxygen concentration and flow velocity, as well as the critical (minimum) mass and heat flux for ignition. Alternatively, a larger sample 100 mm long with a width of 30 mm can be fit in the tunnel. The sample is exposed to a heat flux that decays in the longitudinal direction. The sample is ignited locally at one end, and the rate of flame spread can be measured by digitizing infrared images. The ignition and flame spread data are plotted on a "flammability diagram" which can be used quantify a material's overall flammability. To complement the experimental program, theoretical modeling of the ignition and flame spread process is performed. One of the principal purposes of the modeling effort is to extend the test results to other conditions that have not been tested (different O2 or gravity levels) via modeling. The proposed research program will study the effects of microgravity on the flammability diagrams of solid combustible materials, and the fire properties derived from them. The final objective of the research is to develop a new flammability apparatus that would better reflect the potential ambient conditions of space-based environments, i.e., microgravity, low velocity flow, oxygen concentrations slightly below and above that of air, and use it to obtain more appropriate flammability diagrams for practical materials (plastics, cable jackets, electronic circuit boards, etc.) that will be found in space-based facilities. The strong possibility of an accidental fire occurring in a space-based facility during a long mission, is a primary safety concern, and consequently it is of critical importance to characterize the fire properties of materials that are used.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Composites are finding increased use in the transportation and construction industries. One barrier to their widespread adoption is the increased flammability of polymer based composites when compared with other materials, particularly metals and concrete. In the process of developing a test methodolgy for flammability of composites intended for use in space structures, the FIST project is developing a better understanding of composite flammability useful to ground based design and manufacture.
	During FY05, considerable effort was devoted to developing modeling tools that can be used to calculate the ignition time of an irradiated solid fuel in microgravity environments. Conventionally, a solid-phase pyrolysis model is used to calculate the transient temperature distribution and mass loss rate of an irradiated solid. Ignition is assumed to occur when the mass loss rate reaches a critical value known as the critical mass loss rate at ignition. The primary shortcoming of this approach for predicting ignition times of a solid fuel in microgravity is that it is difficult to estimate how this critical mass loss rate at ignition varies with forced flow velocity, ambient oxygen concentration, heat flux level, and from fuel to fuel. For these reasons, a large number of FIST experiments were conducted to measure the critical mass loss rate at ignition of the polypropylene/glass composite (PP/GL) and polymethylmethacrylate (PMMA) fuels as a function of oxygen concentration, heat flux level, and forced flow velocity. Heat flux levels were varied from 10 to 35 kW/m2, oxygen concentrations from 15-30%, and forced flow velocity from 0.5 m/s to 1.75 m/s. Data was collected on continuous mass loss with precision of +/- 0.05 mg at 0.5 second intervals. Front and backside surface temperature measurements were made with Type K thermocouples at 0.5 second intervals and providing +/- 5% precision. Values of dm/dt were calculated using a linear fit over an interval of 10-15 seconds around the timed point of observed ignition. As a complement to the experimental measurements of the critical mass loss rate at ignition, a simple theory was developed that can be used to predict the critical mass loss rate at ignition as a function of fuel properties (heat of combustion and stoichiometric oxygen to fuel ratio), forced flow velocity, and ambient oxygen concentration. This work was submitted to the 31st International Symposium on Combustion. Good agreement between the experimental measurements and the model predictions were obtained. In addition to measuring the critical mass loss rate at ignition, long-duration mass loss experiments were conducted with the igniter de-energized to help understand how the forced flow velocity and oxygen concentration affect the mass loss rate at a given heat flux. A major part of the theoretical work during FY05 involved development of a comprehensive pyrolysis model to simulate the heating and gasification (i.e. mass loss) processes that occur in combustible solids. The model simulates the relevant physical processes that occur during the gasification and burning of complex solids, particularly charring or intumescent (swelling) materials and composites (due to their widespread use in the aerospace industry). Although the model has been developed with complex materials in mind, it can also be used to simulate simple noncharring solids such as thermoplastics. The governing solid-phase conservation equations describing the pyrolysis of a charring/intumescent material have been formulated and implemented into a numerical model. The numerical model solves conservation equations for a user-specified number of solid and gas-phase species. For example, in simulating the thermal decomposition process in a charring material, conservation equations would be solved for the virgin material, the char, and the oxidized char (ash). Each solid species has its own temperature-dependent thermophysical properties (thermal conductivity, density, specific heat, radiation absorption coefficient, surface emissivity, etc.). The user may also specify an arbitrary number of solid to gas, solid to solid, or solid to solid + gas chemical reactions, each having its own Arrhenius reaction rate. Melting is handled using an apparent specific heat wherein the latent heat of melting is added to the specific heat of the matrix in the vicinity a "melting temperature"; it is assumed to follow a Gaussian distribution. Additional physics incorporated in the model include in-depth radiation absorption according to the Beer/Lambert law, in-depth radiation transport across pores using the "radiant conductivity" approach, gas-solid heat transfer, and Arrhenius reactions having arbitrary reaction order. Due to its generality, the model may be applied to thermoplastic fuels (PMMA, PP), charring fuels such as wood and fire-retarded polymers, intumescent solids, blended composites, laminated composites, melting polymers, non-homogeneous solids having layered composition, etc.

Task Progress:	As part of the model verification process, the temperature profiles predicted by the model have been compared to exact solutions for two cases. The first is a semi-infinite solid heated by a constant incident heat flux with surface losses by convection only. The second is the classical “Stefan problem” which describes the propagation of a solidification front into a semi-infinite medium initially in the liquid state. The model accurately reproduces the temperature profiles for both exact solutions examined. No analytical solutions are available for the cases where pyrolysis occurs or the thermal properties vary with temperature. However, the model has been exercised quite vigorously and integral balances on the solid as a whole have confirmed that the model strictly conserves energy, mass, and species. This indicates that the model equations have been correctly implemented. Due to the level of detail included in the pyrolysis model, a particular material must be characterized by specifying a large number of “material properties”, most of which cannot be directly measured or inferred from existing small-scale fire tests. For this reason, a procedure has been developed to extract the material properties required to characterize a material from existing bench scale fire tests (such as FIST). The methodology works by finding a combination of material properties (thermal properties, reaction kinetics, enthalpies of reaction, etc.) that provide optimal agreement between the model predictions and experimental data, which consists of temperature and mass loss rate measurements for a radiatively heated solid. Since a charring or intumescent material may require specification of more than 20 material properties, conventional search methods are not appropriate due to the large search space and the inherent nonlinearity. For this reason, the optimization procedure uses a genetic algorithm, which uses techniques rooted in “survival of the fittest” or Darwinian evolution to find an optimal solution (set of material properties providing the best agreement between model predictions and experimental data). A paper describing this methodology has recently been accepted for publication in Fire Safety Journal and will appear in 2006. To reduce the required CPU time, the genetic algorithm has been parallelized using Message Passing Interface (MPI) and the code has been successfully run in parallel on UCB computer clusters. The pyrolysis model described above is coupled to NIST’s Fire Dynamics Simulator v4.05, a CFD code geared toward simulation of fire phenomena. The combined pyrolysis model/CFD code has been used to simulate opposed flow flame spread, including the effect of environmental variables on the predicted flame spread rate. It was found that at a fixed velocity the predicted flame spread rate is quite sensitive to the ambient oxygen concentration, increasing with O2 concentration. However, at a fixed oxygen concentration it was found that the predicted flame spread rate is not sensitive to the freestream velocity. This may correspond to the “flat region” seen at relatively low oxygen concentrations in the Fernandez-Pello opposed flame spread curves where the opposed flow flame spread velocity is independent of the freestream velocity. A three-dimensional CFD model of the KC-135 FIST insert has been developed and coupled to the custom pyrolysis model being developed as part of the FIST project. The three dimensional model makes it possible to investigate the effects of the flow enclosure on the ignition and flame spread behavior in reduced gravity environments. The model is being used to better understand the effects of flow velocity, oxygen concentration, and gravity level on opposed flow flame spread rates. A limited number of three-dimensional calculations were performed with an infinitely fast gas-phase combustion reaction to investigate the effects of the flow enclosure. More recently, modeling efforts have been focused on 2D calculations using a finite-rate one step combustion reaction. Computer power limitations prevent 3D finite rate chemistry calculations from being performed. The experimental flame spread data obtained from the October 2004 KC-135 flame spread campaign are being used to “calibrate” the flame spread calculations. The FIST team was trained by a representative from TSI Incorporated on use of the PIV system now temporarily installed on the FIST duct. Images of velocity vectors were obtained but considerable work needs to be done to give useable data on velocity fields near the reaction zone. Analysis of flame spread images from the KC-135 are underway with successful extraction of flame spread velocities using SPOTLIGHT Image Analysis Software. Continuous burning mass loss measurements have begun in order to provide data to heat release rate calculations underway.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
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